YUNUS EMRE ULUÇ

Junior AI Engineer · LLMs, Automation & Python

+90 551 682 9189 • y.emreuluc@gmail.com • github.com/yunusemreuluc • linkedin.com/in/yunusemreuluc • Konya, Türkiye



Summary

Junior AI Engineer and Software Engineering graduate (2026), **valedictorian of both my faculty and my department**. I build practical AI applications in Python: **LLM-powered tools, API integrations and workflow automation**. I prepare and clean data, test AI outputs for accuracy and reliability, and document what I find.

I use AI coding assistants (Claude Code, Codex CLI) every day, and I read, test and debug everything they produce before it ships. Projects below are on GitHub with code, tests and documentation.

AI Projects

Airon — LLM-Powered Desktop Assistant & MCP Server 2026

Sole Developer · Python · FastAPI · WebSocket · Gemini Live API · Next.js · github.com/yunusemreuluc/Airon

- Built an LLM application with **33 tools exposed through function calling** on the Gemini Live API: tool schemas, routing, error handling and multi-step task execution.
- Added an **MCP (Model Context Protocol) server** with 4 tools using the official Python SDK, covered by **42 pytest tests** run through a real MCP client; verified in MCP Inspector and connected to Claude Code.
- Reviewed the server for reliability and safety and fixed 7 of 8 findings (input validation, personal-data masking, resource limits, clearer error messages).
- Integrated AI features into a working product: FastAPI + WebSocket backend, Next.js frontend, on-device object detection (YOLO-World) and OCR, and a two-step user-approval layer for risky actions.

Workflow Automation — Digital Product Pipeline & YouTube Shorts Uploader 2026

Sole Developer · Python · Pillow · ReportLab · YouTube Data API v3 · OAuth

- Automated an image-generation, mockup, listing-copy and PDF-delivery workflow into **a single command**; 39 product sets produced with it.
- Wrote an uploader that publishes generated short videos on a daily schedule through the **YouTube Data API**, handling OAuth credentials and scheduled publishing.

FractureAI — TÜBİTAK 2209-A Funded AI Research Project 2025 – 2026

Project Lead · PyTorch · FastAPI · github.com/yunusemreuluc/xray-fracture-detection

- **Prepared and cleaned ~67,000 X-ray images** from four public datasets; patient-level splitting prevented data leakage, Confident Learning handled label noise.
- Trained and evaluated a model ensemble reaching **91.54% accuracy** (Cohen's κ 0.831, AUC 0.960) and shipped it as a FastAPI web app with Grad-CAM explanations.

Dengue Fever Detection — Data Analysis & Model Comparison 2026

Researcher · scikit-learn · XGBoost · Pandas · under review for publication

- Analysed 1,523 records with 19 parameters and compared nine ML algorithms with Stratified K-Fold cross-validation.

Experience

Software Engineering Intern — Opak Yazılım, Konya 2025

Mobile & Cross-Platform Development · engineering internship · applications delivered and presented

- Built and delivered **NICEO & NEO**, mobile e-commerce applications covering product catalogue, cart and checkout flows, with Flutter/Dart and React Native/TypeScript.
- Built the REST API integration on a UML-modelled layered architecture with explicit state management; presented the completed applications to the company.

Skills

Programming: Python, TypeScript, JavaScript, SQL, Kotlin, C++, C#

AI and LLMs: Large Language Models (LLM), Gemini API, function calling, MCP (Model Context Protocol), prompt engineering, AI output testing and evaluation

Machine Learning: PyTorch, scikit-learn, XGBoost, OpenCV, ONNX Runtime, data cleaning and preparation, Pandas, NumPy

APIs and Automation: REST APIs, FastAPI, WebSockets, API integration, workflow automation, OAuth

Tools: Git and GitHub, pytest, Docker, Linux, Claude Code, Codex CLI, Jupyter, VS Code

Education

B.Sc. Software Engineering — Muş Alparslan University 2022 – 2026

Valedictorian of the Faculty and the Department · graduated with the highest GPA

- Thesis: “A Deep Learning Based AI System for Fracture and Dislocation Detection in Clinical X-Ray Images”.

Certifications and Languages

CCNA: Introduction to Networks — Cisco Networking Academy Jan 2025

Turkish — Native **English** — B1 · technical documentation and academic literature